

1854:

Maximum Population year

Q Constraints: $1950 \leq \text{birth} < \text{Death} \leq 2050$

Example: $[[1993, 2000], [2001, 2008]]$
logs: $[\text{birth}, \text{death}]$

Return earliest year with maximum population

1) Create array of 2051 length & initialized all with 0's
length 2051 b/c index in constraints is 2050

`int[] year = new int[2051];`

2) Use for loop till less than logs length, b/c starts from 0

`for(int i=0; i < logs.length; i++)`

3) now store the first ~~index~~ element of i th row in a variable &

`int birth = logs[i][0]` 0th row 0th index
mean 1993

so birth = 1993

for death year

int death = log[i][1] ^{0th row}
1st index

so death = 2000

now for checks & make our array "year"
indexes from birth year 1993 to 2000

0s to 1 mean now it is 0 so we have
to change the value of ^{from} birth year ^{from 0} to 1
& death year from 0 to -1

year [birth]++

year [death]--

^{index}
year [1993]++

↳ mean go to index

↳ year [2000]--

1993 & change the
value of that

↓
make 2000th
index value
from 0 to -1.

index from 0 to
1

(4) Now: we need 3 variables

sum → to store 1 where person is alive otherwise 0

max → to get & compare if sum > max

so max will store that value

ans $\rightarrow$ to get the earliest year of maximum population

so

```
int sum = 0;
```

```
int max = 0;
```

```
int ans = 1950;
```

$\rightarrow$ earliest possible max year
b/c constraints value starts from 1950

now use for loop, we have to check from

1950 to ^{earliest} 2050

$\rightarrow$ why less than 2050?

b/c Problem says

for (int i = 1950; i < 2050; i++) that do not include the death year of person so death year - 1;

```
sum = sum + year[i];
```

$\rightarrow$ sum = 0 + year[1950]

```
if (sum > max) {
```

```
    max = sum;
```

```
    ans = i;
```

```
}
```

```
return ans; // will return year
```

$\hookrightarrow$ index
So in year array at index 1950 the value is

0. so

sum = 0 + 0

sum = 0
& going on till 1993.

now how it works?

when i will equal to 1993

so

$$\text{sum} = \text{sum} + \text{year}[1993]$$

↳ so at that index
the value is

1 so

$$\text{sum} = 0 + 1$$

$$\text{sum} = 1$$

now

$$\text{sum} = 1$$

if ($\text{sum} > \text{max}$) {

$$\text{max} = \text{sum} // \text{max} = 1$$

$$\text{ans} = i;$$

↳ the index mean i is
equal to 1993 so

$$\text{ans} = 1993$$

$$\text{so ans} = 1993$$

↳ but it don't know
so it will check
till length 2050
but no value will
update b/c it
is the earliest
year with
max of 1

is already update
now now change.

like max is
1 so it is
if we use
if $\text{sum} > \text{max}$

population in
out example.